End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech, CSE-AI/ECE-AI>, Semester:<IV>
(April -May, 2023)

Subject Code: BAI 204	Subject: Optimization Techniques and Decision Making

Time :3 Hours | Maximum Marks :50

Note: Q.1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) What is the most important step in an optimization algorithm? Why?
(b) Discuss one application of optimization in the field of i) Computer Science Engineering, and ii) Electronics Communication and Engineering.
(c) Write a short note on Semi-Definite Programming
(d) What is Graphical Optimization? Discuss its importance in detail.
(e) Discuss similarities between Particle Swarm Optimization algorithm and Ant Colony Optimization algorithm.
(f) State and discuss two real world applications of Genetic algorithms.
(g) What are content-oriented methods?
(h) Discuss advantages and disadvantages of using Decision Trees in Decision Making process.

UNIT-I

Q2 Discuss constraint surface in a hypothetical two-dimensional design space along with a diagram. (10)

Q3 Elaborate – Classification of optimization problems based on nature of the equations involved. (10)

UNIT-II

Q4 Attempt any one of the following – (10)

(a) Solve using graphical method - A factory manufactures two types of items, A and B, each type requiring the use of two machines, an automatic and a hand operated. It takes 4 minutes on the automatic and 6 minutes on hand operated machines to manufacture a package of item A, while it takes 6 minutes on automatic and 3 minutes on the hand operated machines to manufacture a package of item B. Each machine is available for at the most 4 hours on any day.
The manufacturer can sell a package of item A at a profit of Rs. 7 and item B at a profit of Rs 10. Assuming that he can sell all the items he manufactures, how many packages of each type should the factory owner produce in a day in order to maximise his profit? Determine the maximum profit.

(b) Solve using Simplex method - Niki holds two part-time jobs, Job I and Job II. She never wants to work more than a total of 12 hours a week. She has determined that for every hour she works at Job I, she needs 2 hours of preparation time, and for every hour she works at Job II, she needs one hour of preparation time, and she cannot spend more than 16 hours for preparation. If she makes $40 an hour at Job I, and $30 an hour at Job II, how many hours should she work per week at each job to maximize her income?

Q5 What is integer programming problem? Discuss Knapsack problem in reference to integer programming problem. (10)

UNIT-III

Q6 Write the algorithm for Particle Swarm Optimization. Discuss the initialization step in detail. State its advantages, disadvantages and two applications. (10)

PTO

Q7 Can crossover operator be used on the given strings P1 and P2? State and discuss the reason. If yes, using crossover operator, do the following on given two strings – P1 – 1101100100110110, and P2- 1010111000011110 (10)

i) Single Point crossover after 5 point.
ii) Two Point Crossover after 5 point each.
iii) Uniform Crossover

UNIT-IV

Q8 Discuss the steps involved in Decision Making with a diagram. Are Decision Making and Decision Analysis the same? Why or Why not? (10)

Q9 Compare and contrast Group Decision Making methods. (10)